DATA SCIENCE

The King of Serving: Tennis Web Scraping with Selenium

The Power of Selenium, Excel and Pandas

Stephen Fordham
Aug 7, 2019 · 7 min read

Josh Calabrese via unsplash

Introduction

Serving in Tennis often determines the outcome of the match. A break of serve or a hold of serve at a key moment in a set can ultimately decide whether silverware beckons.

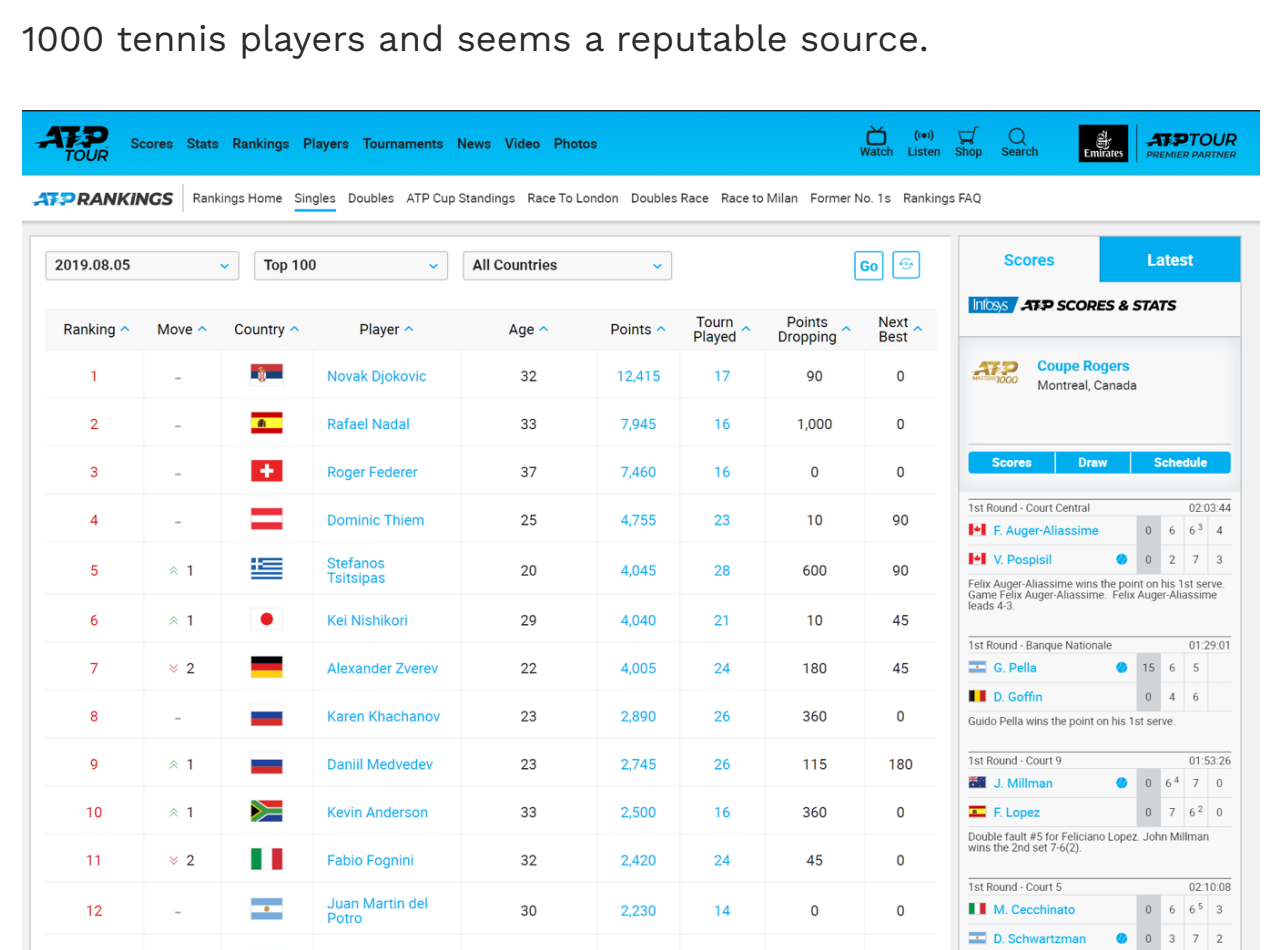
Through the combination of measured web scraping with appropriately placed explicit waits, tennis service data can be extracted, exported to Excel, and filtered accordingly using the Pandas library.

This tutorial piece will demonstrate how, through the deployment of some ingenuity, the scraping script runs smoothly and returns data in a convenient, easy-to-analyse format.

Since the service games of Tennis players can decide the difference between glory on the court or an early tournament exit, I will scrape data on the current Top 20 Men's Tennis players who compete on the ATP circuit. Specifically, I want to gather data which will enable discrimination between the tennis players based on their serving alone.

Finding the Data

To extract the data, I first need to find a source, which is up-to-data. The ATP tour website has comprehensive information on over 1000 tennis players and seems a reputable source.



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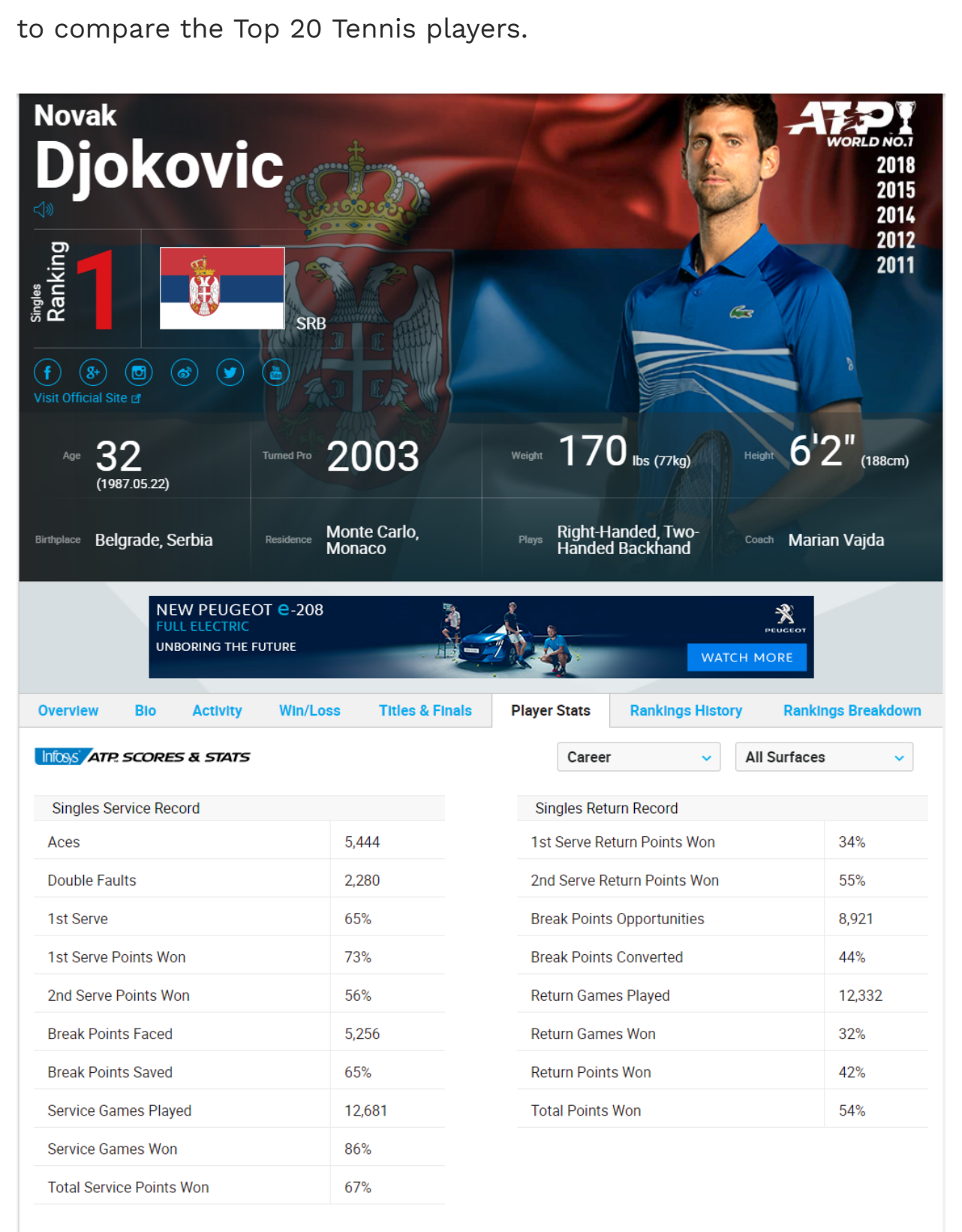
Ranking	Move	Country	Player	Age	Points	Tourn Played	Points Dropping	Next Best
1	-		Novak Djokovic	32	12,415	17	90	0
2	-		Rafael Nadal	33	7,945	16	1,000	0
3	-		Roger Federer	37	7,460	16	0	0
4	-		Dominic Thiem	25	4,755	23	10	90
5	1		Stefanos Tsitsipas	20	4,045	28	600	90
6	1		Kei Nishikori	29	4,040	21	10	45
7	2		Alexander Zverev	22	4,005	24	180	45
8	-		Karen Khachanov	23	2,890	26	360	0
9	1		Daniil Medvedev	23	2,745	26	115	180
10	1		Kevin Anderson	33	2,500	16	360	0
11	2		Fabio Fognini	32	2,420	24	45	0
12	-		Juan Martin del Potro	30	2,230	14	0	0
13	-		Roberto Bautista Agut	31	2,215	23	0	0
14	-		Borna Coric	22	2,195	22	45	0

Most importantly, when the link text for each player is clicked on, serving-specific data can be seen as shown in the bottom half of the image below. It is this information, that can be extracted and used to compare the Top 20 Tennis players.



Age 32 (1987.05.22)
Turned Pro 2003
Weight 170 lbs (77kg)
Height 6'2" (188cm)
Birthplace Belgrade, Serbia
Residence Monte Carlo, Monaco
Plays Right-Handed, Two-Handed Backhand
Coach Marian Vajda

Singles Service Record	
Aces	5,444
Double Faults	2,280
1st Serve	65%
1st Serve Points Won	73%
2nd Serve Points Won	56%
Break Points Faced	5,256
Break Points Saved	65%
Service Games Played	12,681
Service Games Won	86%
Total Service Points Won	67%

Singles Return Record	
1st Serve Return Points Won	34%
2nd Serve Return Points Won	55%
Break Points Opportunities	8,921
Break Points Converted	44%
Return Games Played	12,332
Return Games Won	32%
Return Points Won	42%
Total Points Won	54%

Code Workflow

Before extraction, it is useful to decide on a code workflow. Specifically, I would like to:

Iterate through the Top 20 Men's Tennis players on the ATP circuit, and extract their name, the year they turned pro, and the tabular data inside the 'Singles Service Record' shown above.

Importing Selenium Classes

and tqdm

To begin, I will include various import statements for selenium classes. I need to import the webdriver to navigate to the webpage, the By class for help finding web element locators, the options class to instantiate an options object so I can have a headless browser.

Further to this, I will need to import the WebDriverWait class and the expected_conditions class, which work in tandem. Information pertaining to all of these selenium classes can be found by following this link. Examples of their use will be shown below.

Finally, although *not* strictly necessary, I will import tqdm so I can tract the progress of my script in real-time in the console of my Integrated Development Environment (IDE).

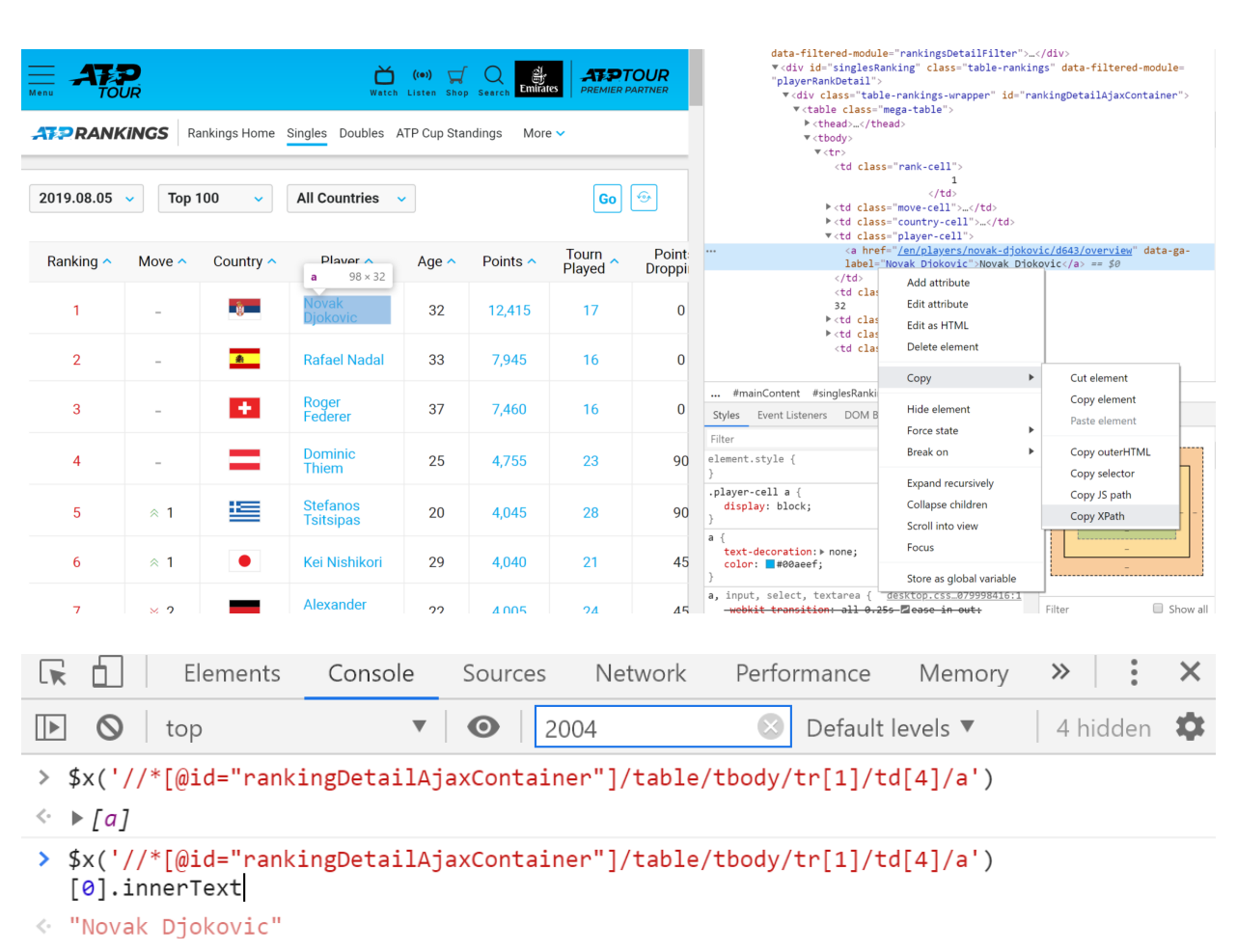
Writing the Extracted Output to Excel

On each iteration, data is transferred to a comma separated values (CSV) file. To begin, I first created a File called 'Tennis_player_details_scrape.csv' and opened this newly created file in write, 'w' mode.

Before I fetch the data and serving-specific information is added to the CSV file, I create some sensibly named headers(column titles) for the CSV file, and include a new-line character.

Finding and Adding Data to the Growing CSV file

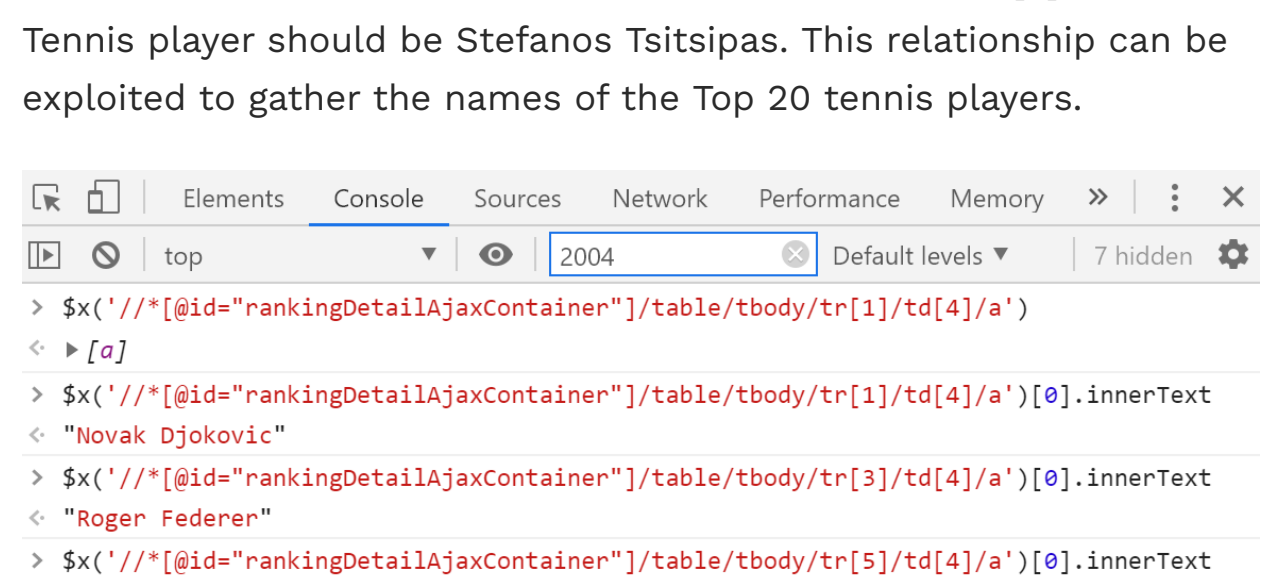
The first objective is to find the first Tennis player. By right-clicking, followed by inspecting, the tennis player can be found within the HTML. This Xpath can be checked that it corresponds to Novak Djokovic using the console tool in the developers tab in Chrome. The Xpath returns a list of one **a link**. Indexing this element (the zeroth element), and extracting the Text, using the innerText function returns the Tennis player Novak Djokovic.



```
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[1]/td[4]/a')
< ▸ [a]
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[1]/td[4]/a')
  [0].innerText
< "Novak Djokovic"
```

Find the Pattern, Find all the Players

The Xpath specific to Novak Djokovic, as shown corresponds to row number 1. By simply changing this number to 3, (tr[3]) for arguments sack, should return Roger Federer. The results in the console confirm this. To further validate; if the table row is 5, tr[5], the Tennis player should be Stefanos Tsitsipas. This relationship can be exploited to gather the names of the Top 20 tennis players.



```
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[1]/td[4]/a')
< ▸ [a]
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[1]/td[4]/a')[0].innerText
< "Novak Djokovic"
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[3]/td[4]/a')[0].innerText
< "Roger Federer"
> $x('//*[@id="rankingDetailAjaxContainer"]/table/tbody/tr[5]/td[4]/a')[0].innerText
< "Stefanos Tsitsipas"
> |
```

This can be achieved by creating a for loop with a range of 1 to 21 and replacing the number with i.

To further strengthen the code and prevent any timeout exceptions, in case the web pages load slowly, I will set an explicit wait of 20 seconds by using the WebDriverWait class. Here the browser and time are required arguments for this class. Furthermore I will wait until the web element is visible, before the text is extracted using the expected conditions class with the visibility_of_element_located method set.

This method takes a locator as a required argument. This is specified using 'By.XPATH, followed by the xpath in quotations.

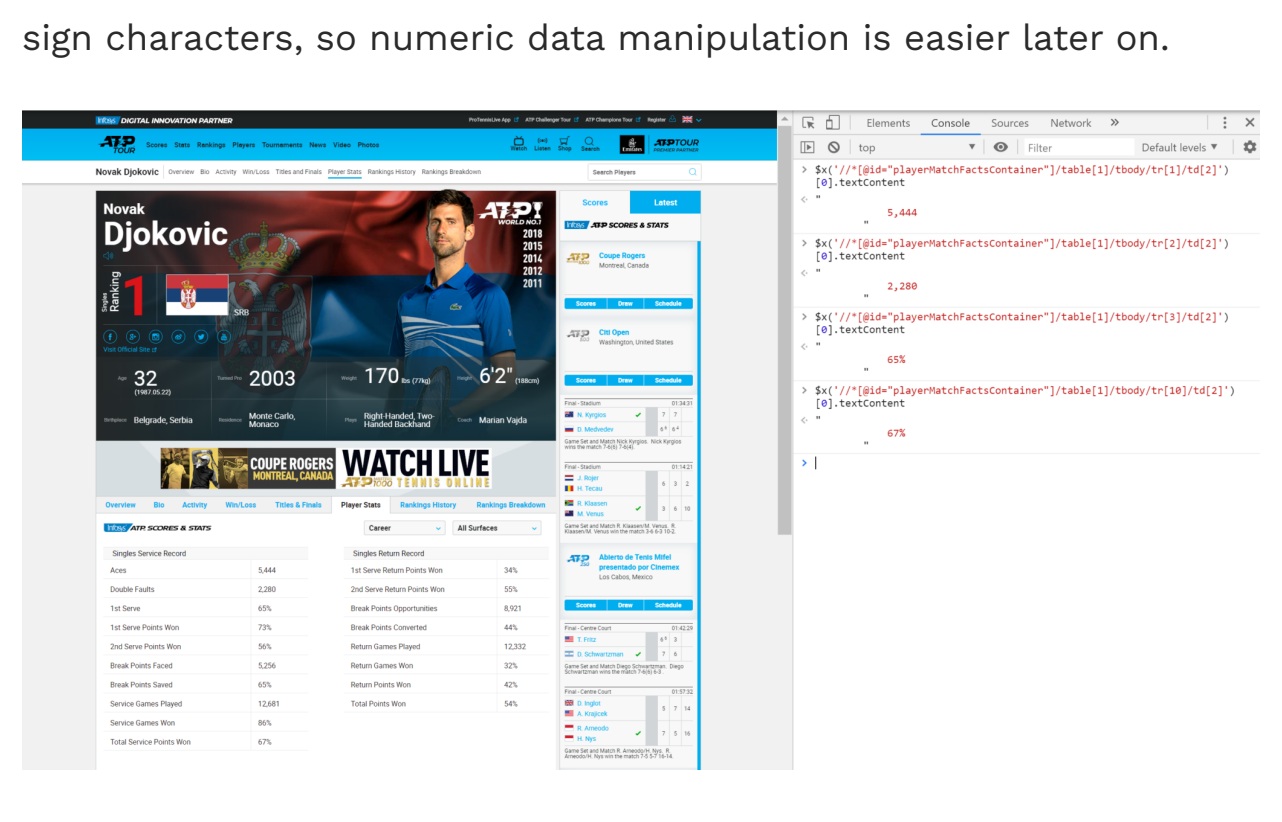
I write the player name to the CSV file and click on the link of each player. I then extract when the player turned pro, and take 2019 from this number to determine the length of their career to date. Again, I write this information to the CSV file.

In all cases, where I want to extract information, I wait until the particular element I am interested in finding is visibility on the web page. This prevents the occurrence of timeout exceptions being raised.

Player Statistics

In order to scrape the player statistics, I click on the Player Stats tab, and data relating to that player's 'Single Service Record' is returned. Scraping this data is simple. Once more, the only portion that changes is the number inside the table row[], for example tr[1] tells me how many Aces Djokovic has served, whereas tr[10] informs me of Djokovic's total service points won.

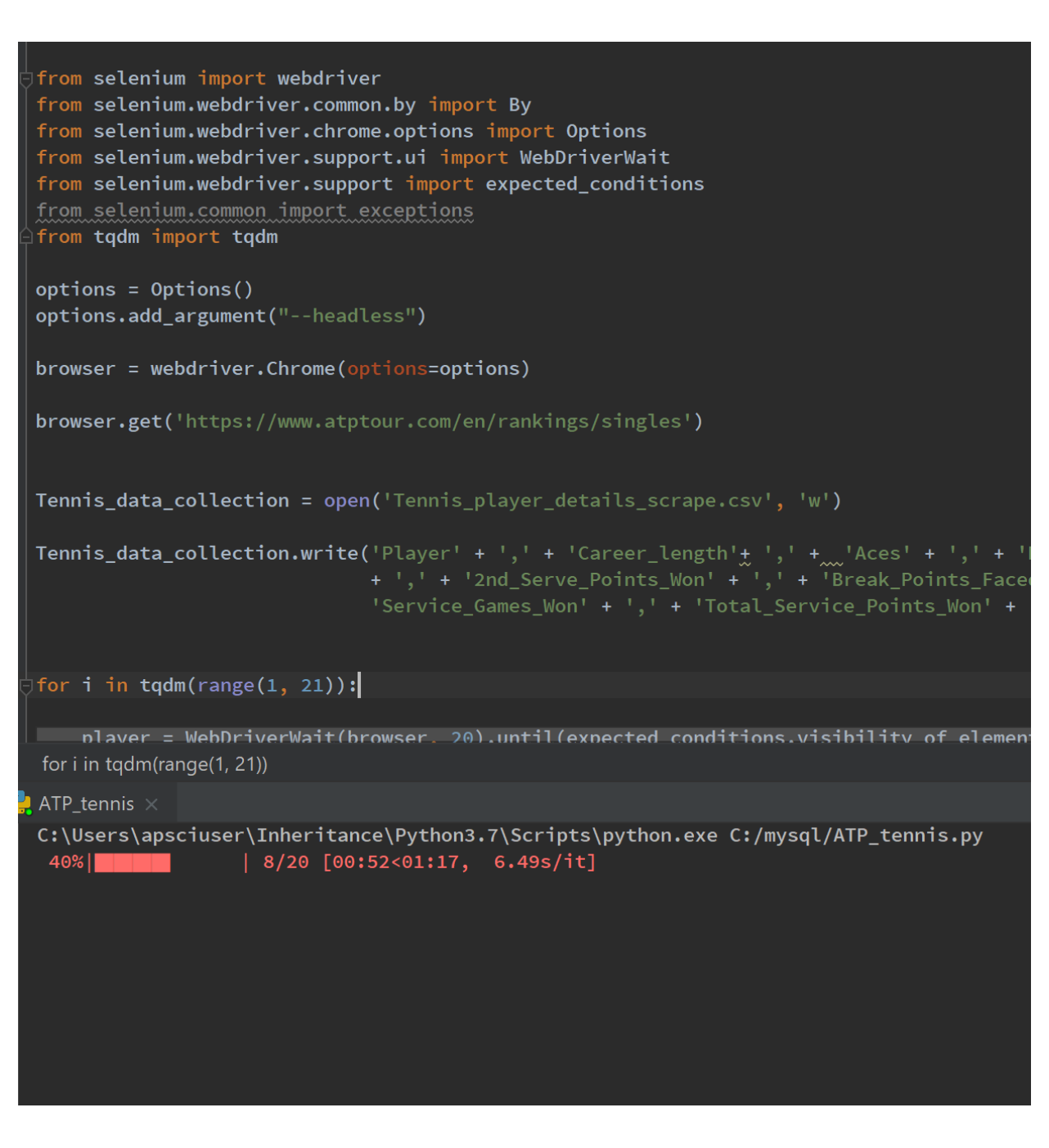
I simply iterate through this organised tabular data and extract all the player statistics, and write them to excel using a nested for loop. To make things easier later on I replace any comma characters or % sign characters, so numeric data manipulation is easier later on.



```
> $x('//*[@id="playerMatchFactsContainer"]/table[1]/tbody/tr[1]/td[2]')
  [0].textContent
< "
        5,444
  "
> $x('//*[@id="playerMatchFactsContainer"]/table[1]/tbody/tr[2]/td[2]')
  [0].textContent
< "
        2,280
  "
> $x('//*[@id="playerMatchFactsContainer"]/table[1]/tbody/tr[3]/td[2]')
  [0].textContent
< "
        65%
  "
> $x('//*[@id="playerMatchFactsContainer"]/table[1]/tbody/tr[10]/td[2]')
  [0].textContent
< "
        67%
  "
> |
```

After this data has been scraped and written to Excel, I navigate back to the singles page to start another iteration.

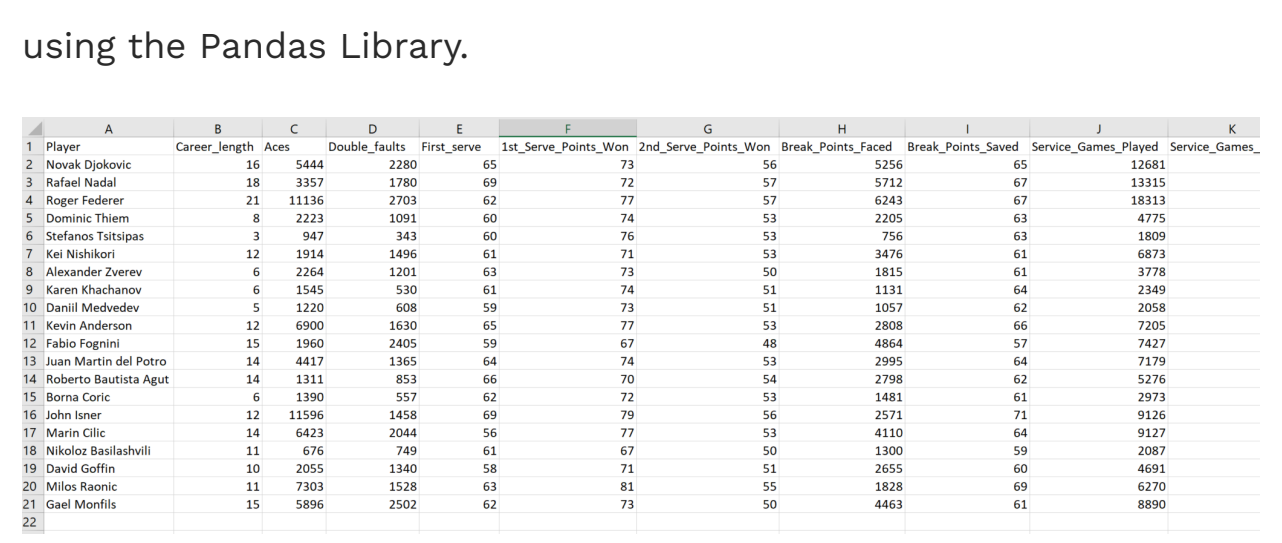
To give me an idea of the progress of the script I import **tqdm**, which prints a dynamically updating progress bar, and wrap this function around my range. In the example below, I can see the progress is 40% and 8 out of 20 iterations are complete with 1:17 minutes remaining.



```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions
from selenium.common import exceptions
from tqdm import tqdm

options = Options()
options.add_argument("--headless")

browser = webdriver.Chrome(options=options)

browser.get('https://www.atptour.com/en/rankings/singles')


Tennis_data_collection = open('Tennis_player_details_scrape.csv', 'w')

Tennis_data_collection.write('Player' + ',' + 'Career_length'+ ',' + 'Aces' + ',' + '
                             + ',' + '2nd_Serve_Points_Won' + ',' + 'Break_Points_Face
                             'Service_Games_Won' + ',' + 'Total_Service_Points_Won' +


for i in tqdm(range(1, 21)):

    player = WebDriverWait(browser, 20).until(expected_conditions.visibility_of_element
```

```
C:\Users\apsciuser\Inheritance\Python3.7\Scripts\python.exe C:/mysql/ATP_tennis.py
 40%|████      | 8/20 [00:52<01:17,  6.49s/it]
```

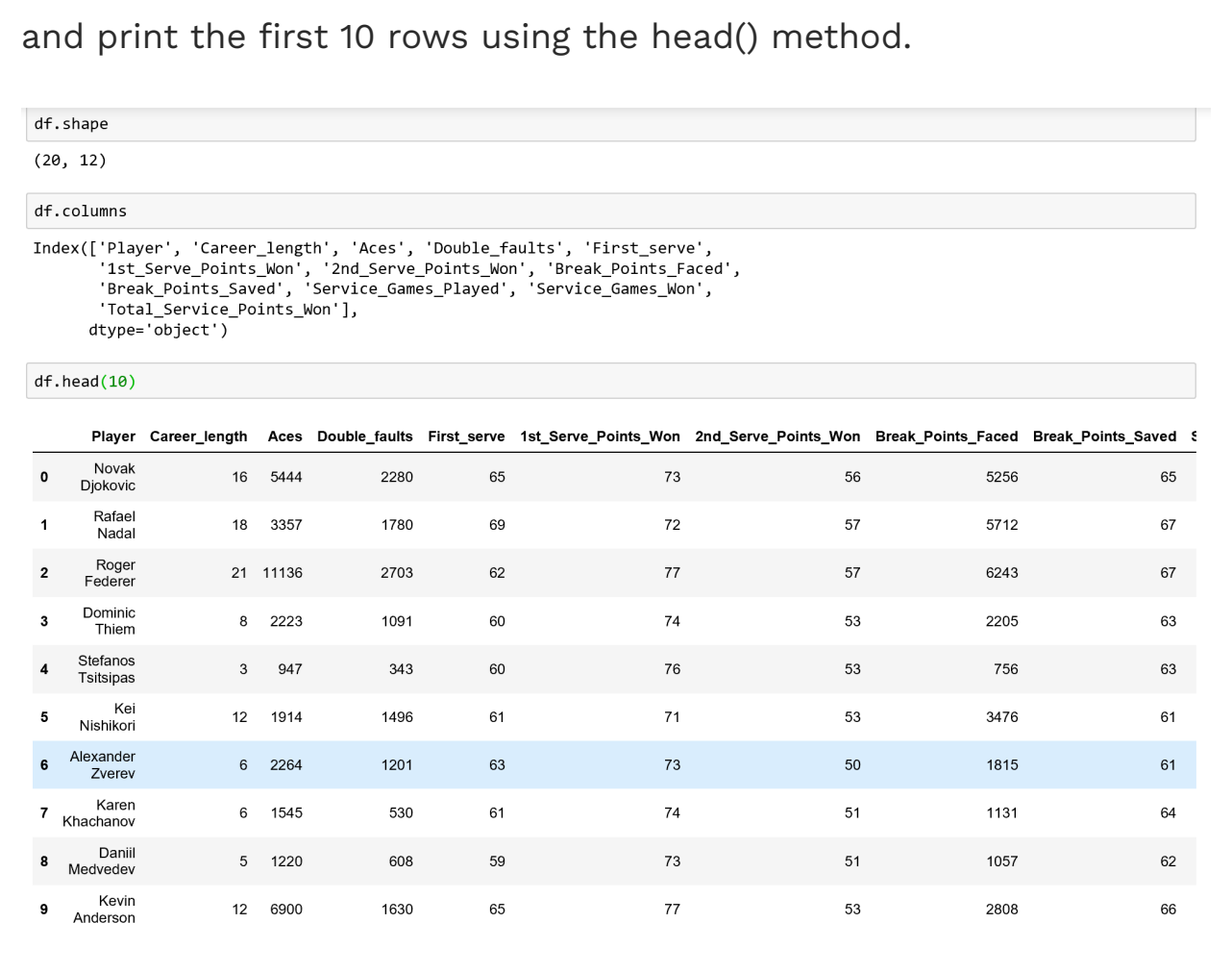
Opening the Tennis_player_details_scrape.csv file in Excel, reveals that all the data has been obtained. Now the data can be filtered using the Pandas Library.



	A	B	C	D	E	F	G	H	I	J
1	Player	Career_length	Aces	Double_faults	First_serve	1st_Serve_Points_Won	2nd_Serve_Points_Won	Break_Points_Faced	Break_Points_Saved	Service_Games_Played
2	Novak Djokovic	16	5444	2280	65	73	56	5256	65	12681
3	Rafael Nadal	18	3357	1780	69	72	57	5712	67	13315
4	Roger Federer	21	11136	2703	62	77	57	6243	67	18313
5	Dominic Thiem	8	2223	1091	60	74	53	2205	63	4775
6	Stefanos Tsitsipas	3	947	343	60	76	53	756	63	1809
7	Kei Nishikori	12	1914	1496	61	71	53	3476	61	6873
8	Alexander Zverev	6	2264	1201	63	73	50	1815	61	3778
9	Karen Khachanov	6	1545	530	61	74	51	1131	64	2349
10	Daniil Medvedev	5	1220	608	59	73	51	1057	62	2058
11	Kevin Anderson	12	6900	1630	65	77	53	2808	66	7205
12	Fabio Fognini	15	1960	2405	59	67	48	4864	57	7427
13	Juan Martin del Potro	14	4417	1365	64	74	53	2995	64	7179
14	Roberto Bautista Agut	14	1311	853	66	70	54	2798	62	5276
15	Borna Coric	6	1390	557	62	72	53	1481	61	2973
16	John Isner	12	11596	1458	69	79	56	2571	71	9126
17	Marin Cilic	14	6423	2044	56	77	53	4110	64	9127
18	Nikoloz Basilashvili	11	676	749	61	67	50	1300	59	2087
19	David Goffin	10	2055	1340	58	71	51	2655	60	4691
20	Milos Raonic	11	7303	1528	63	81	55	1828	69	6270
21	Gael Monfils	15	5896	2502	62	73	50	4463	61	8890
22										

Pandas Filtering of Serving Data

The singles service data is now amendable for analysis. To get started, I first read in the Dataframe, check its shape, and columns and print the first 10 rows using the head() method.



```
df.shape
```

```
(20, 12)
```

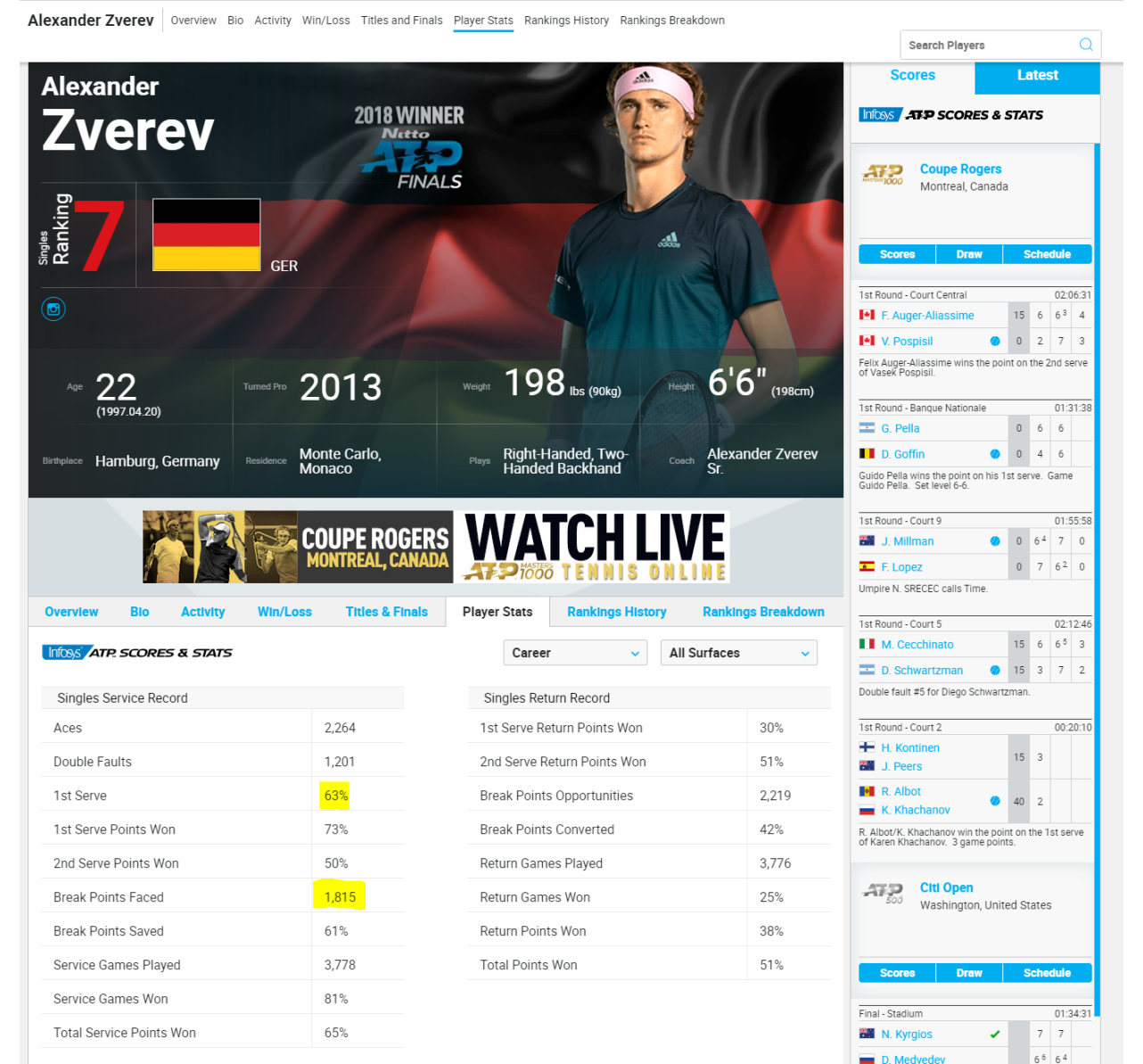
```
df.columns
```

```
Index(['Player', 'Career_length', 'Aces', 'Double_faults', 'First_serve',
       '1st_Serve_Points_Won', '2nd_Serve_Points_Won', 'Break_Points_Faced',
       'Break_Points_Saved', 'Service_Games_Played', 'Service_Games_Won',
       'Total_Service_Points_Won'],
      dtype='object')
```

```
df.head(10)
```

	Player	Career_length	Aces	Double_faults	First_serve	1st_Serve_Points_Won	2nd_Serve_Points_Won	Break_Points_Faced	Break_Points_Saved
0	Novak Djokovic	16	5444	2280	65	73	56	5256	65
1	Rafael Nadal	18	3357	1780	69	72	57	5712	67
2	Roger Federer	21	11136	2703	62	77	57	6243	67
3	Dominic Thiem	8	2223	1091	60	74	53	2205	63
4	Stefanos Tsitsipas	3	947	343	60	76	53	756	63
5	Kei Nishikori	12	1914	1496	61	71	53	3476	61
6	Alexander Zverev	6	2264	1201	63	73	50	1815	61
7	Karen Khachanov	6	1545	530	61	74	51	1131	64
8	Daniil Medvedev	5	1220	608	59	73	51	1057	62
9	Kevin Anderson	12	6900	1630	65	77	53	2808	66

To confirm that all the data is correctly aligned I perform some validation. As an example, lets pick on Alexander Zverev and check his first serve percentage really is 63% and that he has faced 1815 break points so far in his Tennis career. Indeed, this information marries up, as shown in the highlighted sections above and below.



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Age 22 (1997.04.20)
Turned Pro 2013
Weight 198 lbs (90kg)
Height 6'6" (198cm)
Birthplace Hamburg, Germany
Residence Monte Carlo, Monaco
Plays Right-Handed, Two-Handed Backhand
Coach Alexander Zverev Sr.

Singles Service Record	
Aces	2,264
Double Faults	1,201
1st Serve	63%
1st Serve Points Won	73%
2nd Serve Points Won	50%
Break Points Faced	1,815
Break Points Saved	61%
Service Games Played	3,778
Service Games Won	81%
Total Service Points Won	65%

Singles Return Record	
1st Serve Return Points Won	30%
2nd Serve Return Points Won	51%
Break Points Opportunities	2,219
Break Points Converted	42%
Return Games Played	3,776
Return Games Won	25%
Return Points Won	38%
Total Points Won	51%

Sorting by Service Games Won

A key determining factor when deciding who the best server may be, is sorting the Tennis players by Service games won.

To do this, I will take two columns from the scraped Dataframe, named df, create a new column called rankings, and then sort the Tennis players on the value in the column 'Service_Games_won' using the sort_values method.

The Dataframe returned, called Tennis_Serving reveals John Isner has the highest percentage of Service games won despite having a current World ranking of 15. Milos Roanic has the second highest service games won percentage, but sits at position 19 in the World rankings.

World number 1, Novak Djokovic takes 5th position, whilst Roger Federer and David Goffin's respective world rankings match their positions in the Service Games Won percentage table.

Player	Service_Games_Won	Ranking	Position
John Isner	92	15	1
Milos Raonic	91	19	2
Roger Federer	89	3	3
Kevin Anderson	87	10	4
Novak Djokovic	86	1	5
Rafael Nadal	86	2	6
Juan Martin del Potro	85	12	7
Stefanos Tsitsipas	84	5	8
Marin Cilic	84	16	9
Dominic Thiem	83	4	10
Karen Khachanov	83	8	11
Borna Coric	81	14	12
Gael Monfils	81	20	13
Alexander Zverev	81	7	14
Roberto Bautista Agut	80	13	15
Daniil Medvedev	80	9	16
Kei Nishikori	80	6	17
David Goffin	77	18	18
Nikoloz Basilashvili	74	17	19
Fabio Fognini	72	11	20

Serving Index

While the Tennis player with the highest percentage of Service game wins is probably the best determinant of the Men's best server on the ATP circuit, lets use some of the other scraped data and create a simple Serving index

This index will combine and average; the First serve, 1st serve points won, 2nd serve points won, and the Break points saved percentages. I create a new column for this in the original Dataframe, df, then take the player column and the newly created Serving Index column, add a ranking column and then finally sort the Top 20 Tennis players based on this Serving index.

John Isner again sits top of the pile, with World number 1, Novak Djokovic down in 6th position. Clearly, other aspects of Djokovic's game must make him the current ATP World Number 1.

Player	Serving_index	Ranking	Position
John Isner	68.75	15	1
Milos Raonic	67.00	19	2
Rafael Nadal	66.25	2	3
Roger Federer	65.75	3	4
Kevin Anderson	65.25	10	5
Novak Djokovic	64.75	1	6
Juan Martin del Potro	63.75	12	7
Stefanos Tsitsipas	63.00	5	8
Roberto Bautista Agut	63.00	13	9
Dominic Thiem	62.50	4	10
Karen Khachanov	62.50	8	11
Marin Cilic	62.50	16	12
Borna Coric	62.00	14	13
Alexander Zverev	61.75	7	14
Gael Monfils	61.50	20	15
Kei Nishikori	61.50	6	16
Daniil Medvedev	61.25	9	17
David Goffin	60.00	18	18
Nikoloz Basilashvili	59.25	17	19
Fabio Fognini	57.75	11	20

Summary

This scraping tutorial has demonstrated how scraped data can be tabulated into a Dataframe amenable to querying using the Pandas Library. Here, insight into the best Tennis servers in the Men's game can be identified.

• • •

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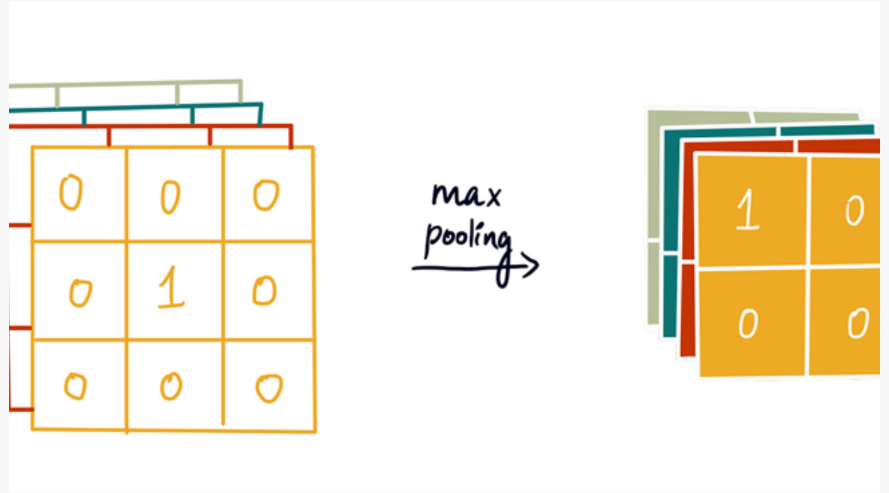

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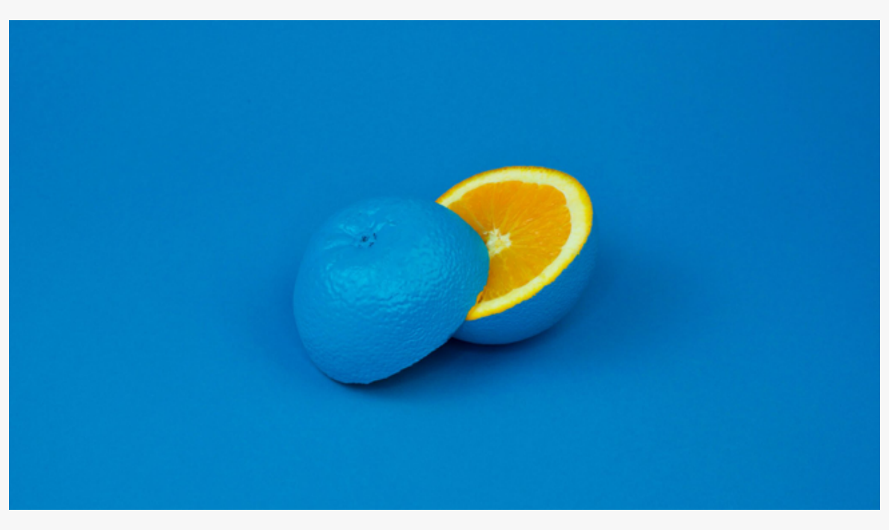

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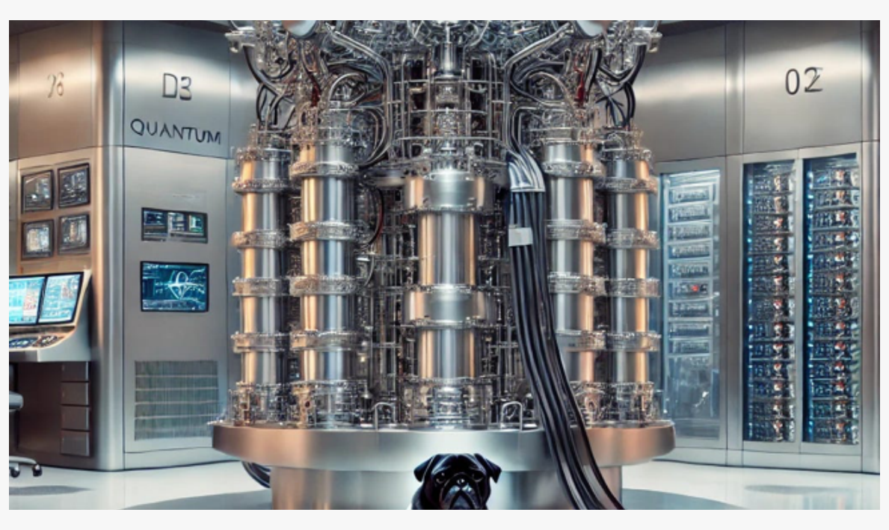

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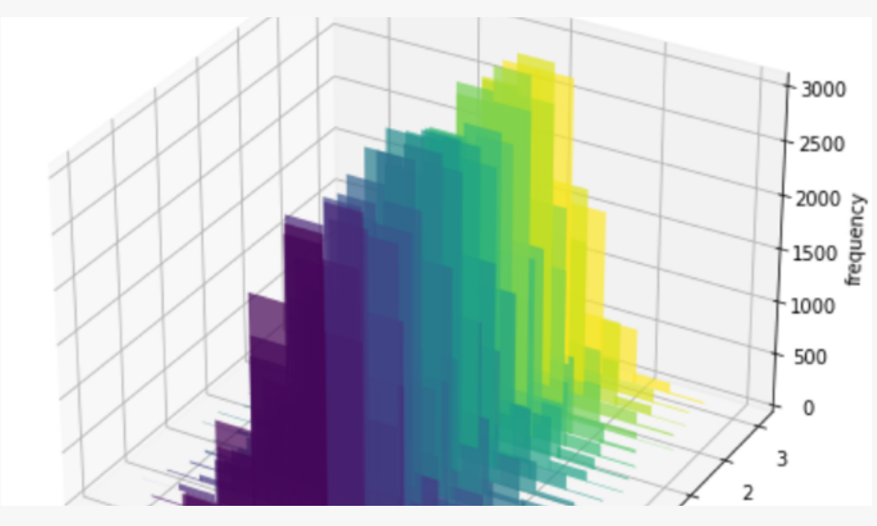

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